FORMALIZED MATHEMATICS
Vol. 24, No. 4, Pages 301–308, 2016
DOI: 10.1515/forma-2016-0026

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Niven's Theorem[1]

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Summary. This article formalizes the proof of Niven's theorem [12] which states that if x/π and $sin(x)$ are both rational, then the sine takes values 0, $\pm 1/2$, and ± 1. The main part of the formalization follows the informal proof presented at Pr∞fWiki (`https://proofwiki.org/wiki/Niven's_Theorem#Source_of_Name`). For this proof, we have also formalized the rational and integral root theorems setting constraints on solutions of polynomial equations with integer coefficients [8, 9].

MSC: 97G60 12D10 03B35

Keywords: Niven's theorem; rational root theorem; integral root theorem

MML identifier: `NIVEN`, version: `8.1.05 5.39.1282`

From now on r, t denote real numbers, i denotes an integer, k, n denote natural numbers, p denotes a polynomial over $\mathbb{R}_{\mathrm{F}}$, e denotes an element of $\mathbb{R}_{\mathrm{F}}$, L denotes a non empty zero structure, and z, z_0, z_1, z_2 denote elements of L.

Now we state the propositions:

(1) Let us consider complexes a, b, c, d. If $b \neq 0$ and $\frac{a}{b} = \frac{c}{d}$, then $a = \frac{b \cdot c}{d}$.

(2) Let us consider real numbers a, b. If $|a| = b$, then $a = b$ or $a = -b$.

(3) If $|i| \leqslant 2$, then $i = -2$ or $i = -1$ or $i = 0$ or $i = 1$ or $i = 2$. The theorem is a consequence of (2).

(4) If $n \neq 0$, then $i \mid i^n$.

(5) If $t > 0$, then there exists i such that $t \cdot i \leqslant r \leqslant t \cdot (i + 1)$.

[1]The work on the formalization presented in this article was completed thanks to the Mizar Mathematical Library maintenance and refactoring carried out at the Computer Center of the University of Białystok.

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ISSN 1426–2630(Print), 1898-9934(Online)

PROOF: Define $\mathcal{P}[\text{integer}] \equiv t \cdot \$_1 \leqslant r$. There exists an integer i_1 such that $\mathcal{P}[i_1]$. Set $F = \lceil \frac{r}{t} \rceil$. For every integer i_1 such that $\mathcal{P}[i_1]$ holds $i_1 \leqslant F$. Consider i such that $\mathcal{P}[i]$ and for every integer i_1 such that $\mathcal{P}[i_1]$ holds $i_1 \leqslant i$ from [15, Sch. 6]. □

(6) Let us consider a finite sequence p of elements of $\mathbb{R}_{\mathrm{F}}$, and a real-valued finite sequence q. If $p = q$, then $\sum p = \sum q$.
PROOF: Define $\mathcal{P}[\text{finite sequence}] \equiv$ for every finite sequence p of elements of $\mathbb{R}_{\mathrm{F}}$ for every real-valued finite sequence q such that $p = q$ and $p = \$_1$ holds $\sum p = \sum q$. $\mathcal{P}[\emptyset]$ by [16, (43)], [4, (72)]. For every finite sequence f and for every object x such that $\mathcal{P}[f]$ holds $\mathcal{P}[f \frown \langle x \rangle]$ by [2, (36), (38)], [5, (31)], [16, (41), (44)]. For every finite sequence f, $\mathcal{P}[f]$ from [2, Sch. 3]. □

(7) Let us consider a natural number i, and an element r of $\mathbb{R}_{\mathrm{F}}$. Then $\mathrm{power}_{\mathbb{R}_{\mathrm{F}}}(r, i) = r^i$.
PROOF: Define $\mathcal{P}[\text{natural number}] \equiv \mathrm{power}_{\mathbb{R}_{\mathrm{F}}}(r, \$_1) = r^{\$_1}$. For every natural number n, $\mathcal{P}[n]$ from [1, Sch. 2]. □

(8) $\sin(\frac{5 \cdot \pi}{6}) = \frac{1}{2}$.

(9) $\sin(\frac{5 \cdot \pi}{6} + 2 \cdot \pi \cdot i) = \frac{1}{2}$.

(10) $\sin(\frac{7 \cdot \pi}{6}) = -\frac{1}{2}$.

(11) $\sin(\frac{7 \cdot \pi}{6} + 2 \cdot \pi \cdot i) = -\frac{1}{2}$.

(12) $\sin(\frac{11 \cdot \pi}{6}) = -\frac{1}{2}$.

(13) $\sin(\frac{11 \cdot \pi}{6} + 2 \cdot \pi \cdot i) = -\frac{1}{2}$.

(14) $\cos(\frac{4 \cdot \pi}{3}) = -\frac{1}{2}$.

(15) $\cos(\frac{4 \cdot \pi}{3} + 2 \cdot \pi \cdot i) = -\frac{1}{2}$.

(16) $\cos(\frac{5 \cdot \pi}{3}) = \frac{1}{2}$.

(17) $\cos(\frac{5 \cdot \pi}{3} + 2 \cdot \pi \cdot i) = \frac{1}{2}$.

(18) If $0 \leqslant r \leqslant \frac{\pi}{2}$ and $\cos r = \frac{1}{2}$, then $r = \frac{\pi}{3}$.

(19) Let us consider an add-associative, right zeroed, right complementable, left distributive, non empty double loop structure L, and a sequence p of L. Then $\mathbf{0}.L * p = \mathbf{0}.L$.

Let us consider L, z, and n. One can verify that $\mathbf{0}.L + \cdot (n, z)$ is finite-Support as a sequence of L.

Let us consider a polynomial p over L. Now we state the propositions:

(20) If $z \neq 0_L$, then if $p = \mathbf{0}.L + \cdot (n, z)$, then $\mathrm{len}\, p = n + 1$.
PROOF: the length of p is at most $n + 1$ by [1, (13)], [3, (32)], [14, (7)]. For every natural number m such that the length of p is at most m holds $n + 1 \leqslant m$ by [14, (13)], [3, (31)], [1, (13)]. □

(21) If $z \neq 0_L$, then if $p = \mathbf{0}.L +\cdot (n, z)$, then $\deg p = n$. The theorem is a consequence of (20).

Note that $\mathbf{0}.\mathbb{R}_{\mathrm{F}}$ is $\mathbb{Z}$-valued and $\mathbf{1}.\mathbb{R}_{\mathrm{F}}$ is $\mathbb{Z}$-valued and there exists an element of $\mathbb{R}_{\mathrm{F}}$ which is integer.

Now we state the proposition:

(22) $\mathrm{rng}\langle z\rangle = \{z, 0_L\}$.

PROOF: Set $p = \langle z\rangle$. $\mathrm{rng}\, p \subseteq \{z, 0_L\}$ by [11, (32)], [1, (14)]. $\square$

Let us consider L, z_0, z_1, and z_2. The functor $\langle z_0, z_1, z_2\rangle$ yielding a sequence of L is defined by the term

(Def. 1) $((\mathbf{0}.L +\cdot (0, z_0)) +\cdot (1, z_1)) +\cdot (2, z_2)$.

Now we state the propositions:

(23) $\langle z_0, z_1, z_2\rangle(0) = z_0$.

(24) $\langle z_0, z_1, z_2\rangle(1) = z_1$.

(25) $\langle z_0, z_1, z_2\rangle(2) = z_2$.

(26) If $3 \leqslant n$, then $\langle z_0, z_1, z_2\rangle(n) = 0_L$.

Let us consider L, z_0, z_1, and z_2. Let us observe that $\langle z_0, z_1, z_2\rangle$ is finite-Support.

Now we state the propositions:

(27) $\mathrm{len}\langle z_0, z_1, z_2\rangle \leqslant 3$. The theorem is a consequence of (26).

(28) If $z_2 \neq 0_L$, then $\mathrm{len}\langle z_0, z_1, z_2\rangle = 3$. The theorem is a consequence of (25) and (26).

(29) Let us consider a right zeroed, non empty additive loop structure L, and elements z_0, z_1 of L. Then $\langle z_0\rangle + \langle z_1\rangle = \langle z_0 + z_1\rangle$.

(30) Let us consider a right zeroed, non empty additive loop structure L, and elements z_0, z_1, z_2, z_3 of L. Then $\langle z_0, z_1\rangle + \langle z_2, z_3\rangle = \langle z_0 + z_2, z_1 + z_3\rangle$.

(31) Let us consider a right zeroed, non empty additive loop structure L, and elements z_0, z_1, z_2, z_3, z_4, z_5 of L. Then $\langle z_0, z_1, z_2\rangle + \langle z_3, z_4, z_5\rangle = \langle z_0 + z_3, z_1 + z_4, z_2 + z_5\rangle$. The theorem is a consequence of (23), (24), (25), and (26).

(32) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and an element z_0 of L. Then $-\langle z_0\rangle = \langle -z_0\rangle$.

(33) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and elements z_0, z_1 of L. Then $-\langle z_0, z_1\rangle = \langle -z_0, -z_1\rangle$.

(34) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and elements z_0, z_1, z_2 of L. Then

$-\langle z_0, z_1, z_2\rangle = \langle -z_0, -z_1, -z_2\rangle$. The theorem is a consequence of (23), (24), (25), and (26).

(35) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and elements z_0, z_1 of L. Then $\langle z_0\rangle - \langle z_1\rangle = \langle z_0 - z_1\rangle$. The theorem is a consequence of (32) and (29).

(36) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and elements z_0, z_1, z_2, z_3 of L. Then $\langle z_0, z_1\rangle - \langle z_2, z_3\rangle = \langle z_0 - z_2, z_1 - z_3\rangle$. The theorem is a consequence of (33) and (30).

(37) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and elements z_0, z_1, z_2, z_3, z_4, z_5 of L. Then $\langle z_0, z_1, z_2\rangle - \langle z_3, z_4, z_5\rangle = \langle z_0 - z_3, z_1 - z_4, z_2 - z_5\rangle$. The theorem is a consequence of (34) and (31).

(38) Let us consider an add-associative, right zeroed, right complementable, left distributive, unital, associative, non empty double loop structure L, and elements z_0, z_1, z_2, x of L. Then $\mathrm{eval}(\langle z_0, z_1, z_2\rangle, x) = z_0 + z_1 \cdot x + z_2 \cdot x \cdot x$. The theorem is a consequence of (23), (24), (27), and (25).

Let a be an integer element of $\mathbb{R}_{\mathrm{F}}$. Note that $\langle a\rangle$ is $\mathbb{Z}$-valued.

Let a, b be integer elements of $\mathbb{R}_{\mathrm{F}}$. One can verify that $\langle a, b\rangle$ is $\mathbb{Z}$-valued.

Let a, b, c be integer elements of $\mathbb{R}_{\mathrm{F}}$. Observe that $\langle a, b, c\rangle$ is $\mathbb{Z}$-valued and there exists a polynomial over $\mathbb{R}_{\mathrm{F}}$ which is monic and $\mathbb{Z}$-valued and there exists a finite sequence of elements of $\mathbb{R}_{\mathrm{F}}$ which is $\mathbb{Z}$-valued.

Let F be a $\mathbb{Z}$-valued finite sequence of elements of $\mathbb{R}_{\mathrm{F}}$. One can check that $\sum F$ is integer.

Let f be a $\mathbb{Z}$-valued sequence of $\mathbb{R}_{\mathrm{F}}$. Let us note that $-f$ is $\mathbb{Z}$-valued.

Let g be a $\mathbb{Z}$-valued sequence of $\mathbb{R}_{\mathrm{F}}$. Observe that $f + g$ is $\mathbb{Z}$-valued and $f - g$ is $\mathbb{Z}$-valued and $f * g$ is $\mathbb{Z}$-valued.

Now we state the proposition:

(39) Let us consider a non degenerated, non empty double loop structure L, and an element z of L. Then $\mathrm{LC}\langle z, 1_L\rangle = 1_L$.

Let L be a non degenerated, non empty double loop structure and z be an element of L. One can check that $\langle z, 1_L\rangle$ is monic.

Now we state the proposition:

(40) Let us consider a non degenerated, non empty double loop structure L, and elements z_1, z_2 of L. Then $\mathrm{LC}\langle z_1, z_2, 1_L\rangle = 1_L$. The theorem is a consequence of (28) and (25).

Let L be a non degenerated, non empty double loop structure and z_1, z_2 be elements of L. Let us observe that $\langle z_1, z_2, 1_L\rangle$ is monic.

Let p be a $\mathbb{Z}$-valued polynomial over $\mathbb{R}_{\mathrm{F}}$. Let us note that $\mathrm{LC}\, p$ is integer.

Now we state the proposition:

(41) Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L, and a polynomial p over L. Then $\deg(-p) = \deg p$.

Let us consider an add-associative, right zeroed, right complementable, non empty additive loop structure L and polynomials p, q over L. Now we state the propositions:

(42) If $\deg p > \deg q$, then $\deg(p+q) = \deg p$.

(43) If $\deg p > \deg q$, then $\deg(p-q) = \deg p$.

(44) If $\deg p < \deg q$, then $\deg(p-q) = \deg q$.

(45) Let us consider an add-associative, right zeroed, right complementable, distributive, non degenerated double loop structure L, and a polynomial p over L. Then $\mathrm{LC}\, p = -\mathrm{LC}(-p)$.

(46) Let us consider an add-associative, right zeroed, right complementable, associative, commutative, well unital, almost left invertible, distributive, non degenerated double loop structure L, and polynomials p, q over L. Then $\mathrm{LC}(p * q) = \mathrm{LC}\, p \cdot \mathrm{LC}\, q$. The theorem is a consequence of (19).

Let us consider an add-associative, right zeroed, right complementable, distributive, non degenerated double loop structure L, a monic polynomial p over L, and a polynomial q over L. Now we state the propositions:

(47) If $\deg p > \deg q$, then $p + q$ is monic. The theorem is a consequence of (42).

(48) If $\deg p > \deg q$, then $p - q$ is monic. The theorem is a consequence of (43).

Let L be an add-associative, right zeroed, right complementable, associative, commutative, well unital, almost left invertible, distributive, non degenerated double loop structure and p, q be monic polynomials over L. Let us note that $p * q$ is monic.

Now we state the propositions:

(49) Let us consider an Abelian, add-associative, right zeroed, right complementable, unital, distributive, non empty double loop structure L, elements z_1, z_2 of L, and a polynomial p over L. Suppose $\mathrm{eval}(p, z_1) = z_2$. Then $\mathrm{eval}(p - \langle z_2 \rangle, z_1) = 0_L$.

(50) RATIONAL ROOT THEOREM:
Let us consider a $\mathbb{Z}$-valued polynomial p over $\mathbb{R}_{\mathrm{F}}$, and an element e of $\mathbb{R}_{\mathrm{F}}$. Suppose e is a root of p. Let us consider integers k, l. Suppose $l \neq 0$ and $e = \frac{k}{l}$ and k and l are relatively prime. Then

(i) $k \mid p(0)$, and

(ii) $l \mid \mathrm{LC}\, p$.

The theorem is a consequence of (7), (6), and (4).

(51) Integral root theorem:
Let us consider a monic, $\mathbb{Z}$-valued polynomial p over $\mathbb{R}_{\mathrm{F}}$, and a rational element e of $\mathbb{R}_{\mathrm{F}}$. If e is a root of p, then e is integer. The theorem is a consequence of (50).

(52) Suppose $1 \leqslant n$ and $e = 2 \cdot \cos t$. Then there exists a monic, $\mathbb{Z}$-valued polynomial p over $\mathbb{R}_{\mathrm{F}}$ such that

(i) $\mathrm{eval}(p, e) = 2 \cdot \cos(n \cdot t)$, and

(ii) $\deg p = n$, and

(iii) if $n = 1$, then $p = \langle 0_{\mathbb{R}_{\mathrm{F}}}, 1_{\mathbb{R}_{\mathrm{F}}} \rangle$, and

(iv) if $n = 2$, then there exists an element r of $\mathbb{R}_{\mathrm{F}}$ such that $r = -2$ and $p = \langle r, 0_{\mathbb{R}_{\mathrm{F}}}, 1_{\mathbb{R}_{\mathrm{F}}} \rangle$.

Proof: Define $\mathcal{P}[\text{natural number}] \equiv$ if $1 \leqslant \$_1$, then there exists a monic, $\mathbb{Z}$-valued polynomial p over $\mathbb{R}_{\mathrm{F}}$ such that $\mathrm{eval}(p, e) = 2 \cdot \cos(\$_1 \cdot t)$ and $\deg p = \$_1$ and if $\$_1 = 1$, then $p = \langle 0_{\mathbb{R}_{\mathrm{F}}}, 1_{\mathbb{R}_{\mathrm{F}}} \rangle$ and if $\$_1 = 2$, then there exists an element r of $\mathbb{R}_{\mathrm{F}}$ such that $r = -2$ and $p = \langle r, 0_{\mathbb{R}_{\mathrm{F}}}, 1_{\mathbb{R}_{\mathrm{F}}} \rangle$. $\mathcal{P}[1]$ by [11, (48), (40)]. $\mathcal{P}[2]$ by [6, (7)], (38), (28). For every non zero natural number k such that $\mathcal{P}[k]$ and $\mathcal{P}[k+1]$ holds $\mathcal{P}[k+2]$ by [1, (13)], [13, (38)], (48), [10, (24)]. For every non zero natural number k, $\mathcal{P}[k]$ from [7, Sch. 1]. $\square$

(53) If $0 \leqslant r \leqslant \frac{\pi}{2}$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational, then $r \in \{0, \frac{\pi}{3}, \frac{\pi}{2}\}$. The theorem is a consequence of (52), (1), (49), (48), (51), (3), and (18).

(54) Suppose $2 \cdot \pi \cdot i \leqslant r \leqslant \frac{\pi}{2} + 2 \cdot \pi \cdot i$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational. Then $r \in \{2 \cdot \pi \cdot i, \frac{\pi}{3} + 2 \cdot \pi \cdot i, \frac{\pi}{2} + 2 \cdot \pi \cdot i\}$. The theorem is a consequence of (53).

(55) If $\frac{\pi}{2} \leqslant r \leqslant \pi$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational, then $r \in \{\frac{\pi}{2}, \frac{2 \cdot \pi}{3}, \pi\}$. The theorem is a consequence of (53).

(56) Suppose $\frac{\pi}{2} + 2 \cdot \pi \cdot i \leqslant r \leqslant \pi + 2 \cdot \pi \cdot i$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational. Then $r \in \{\frac{\pi}{2} + 2 \cdot \pi \cdot i, \frac{2 \cdot \pi}{3} + 2 \cdot \pi \cdot i, \pi + 2 \cdot \pi \cdot i\}$. The theorem is a consequence of (55).

(57) Suppose $\pi \leqslant r \leqslant \frac{3 \cdot \pi}{2}$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational. Then $r \in \{\pi, \frac{4 \cdot \pi}{3}, \frac{3 \cdot \pi}{2}\}$. The theorem is a consequence of (53).

(58) Suppose $\pi + 2 \cdot \pi \cdot i \leqslant r \leqslant \frac{3 \cdot \pi}{2} + 2 \cdot \pi \cdot i$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational. Then $r \in \{\pi + 2 \cdot \pi \cdot i, \frac{4 \cdot \pi}{3} + 2 \cdot \pi \cdot i, \frac{3 \cdot \pi}{2} + 2 \cdot \pi \cdot i\}$. The theorem is a consequence of (57).

(59) Suppose $\frac{3 \cdot \pi}{2} \leqslant r \leqslant 2 \cdot \pi$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational. Then $r \in \{\frac{3 \cdot \pi}{2}, \frac{5 \cdot \pi}{3}, 2 \cdot \pi\}$. The theorem is a consequence of (53).

(60) Suppose $\frac{3\cdot\pi}{2} + 2\cdot\pi\cdot i \leqslant r \leqslant 2\cdot\pi + 2\cdot\pi\cdot i$ and $\frac{r}{\pi}$ is rational and $\cos r$ is rational. Then $r \in \{\frac{3\cdot\pi}{2} + 2\cdot\pi\cdot i, \frac{5\cdot\pi}{3} + 2\cdot\pi\cdot i, 2\cdot\pi + 2\cdot\pi\cdot i\}$. The theorem is a consequence of (59).

(61) If $\frac{r}{\pi}$ is rational and $\cos r$ is rational, then $\cos r \in \{0, 1, -1, \frac{1}{2}, -\frac{1}{2}\}$.

(62) If $0 \leqslant r \leqslant \frac{\pi}{2}$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational, then $r \in \{0, \frac{\pi}{6}, \frac{\pi}{2}\}$. The theorem is a consequence of (53).

(63) Suppose $2\cdot\pi\cdot i \leqslant r \leqslant \frac{\pi}{2} + 2\cdot\pi\cdot i$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational. Then $r \in \{2\cdot\pi\cdot i, \frac{\pi}{6} + 2\cdot\pi\cdot i, \frac{\pi}{2} + 2\cdot\pi\cdot i\}$. The theorem is a consequence of (62).

(64) If $\frac{\pi}{2} \leqslant r \leqslant \pi$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational, then $r \in \{\frac{\pi}{2}, \frac{5\cdot\pi}{6}, \pi\}$. The theorem is a consequence of (62).

(65) Suppose $\frac{\pi}{2} + 2\cdot\pi\cdot i \leqslant r \leqslant \pi + 2\cdot\pi\cdot i$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational. Then $r \in \{\frac{\pi}{2} + 2\cdot\pi\cdot i, \frac{5\cdot\pi}{6} + 2\cdot\pi\cdot i, \pi + 2\cdot\pi\cdot i\}$. The theorem is a consequence of (64).

(66) Suppose $\pi \leqslant r \leqslant \frac{3\cdot\pi}{2}$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational. Then $r \in \{\pi, \frac{7\cdot\pi}{6}, \frac{3\cdot\pi}{2}\}$. The theorem is a consequence of (62).

(67) Suppose $\pi + 2\cdot\pi\cdot i \leqslant r \leqslant \frac{3\cdot\pi}{2} + 2\cdot\pi\cdot i$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational. Then $r \in \{\pi + 2\cdot\pi\cdot i, \frac{7\cdot\pi}{6} + 2\cdot\pi\cdot i, \frac{3\cdot\pi}{2} + 2\cdot\pi\cdot i\}$. The theorem is a consequence of (66).

(68) Suppose $\frac{3\cdot\pi}{2} \leqslant r \leqslant 2\cdot\pi$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational. Then $r \in \{\frac{3\cdot\pi}{2}, \frac{11\cdot\pi}{6}, 2\cdot\pi\}$. The theorem is a consequence of (62).

(69) Suppose $\frac{3\cdot\pi}{2} + 2\cdot\pi\cdot i \leqslant r \leqslant 2\cdot\pi + 2\cdot\pi\cdot i$ and $\frac{r}{\pi}$ is rational and $\sin r$ is rational. Then $r \in \{\frac{3\cdot\pi}{2} + 2\cdot\pi\cdot i, \frac{11\cdot\pi}{6} + 2\cdot\pi\cdot i, 2\cdot\pi + 2\cdot\pi\cdot i\}$. The theorem is a consequence of (68).

(70) NIVEN'S THEOREM:
If $\frac{r}{\pi}$ is rational and $\sin r$ is rational, then $\sin r \in \{0, 1, -1, \frac{1}{2}, -\frac{1}{2}\}$.

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Received December 15, 2016